



Vision of Air

Hackathon	ET AI Hackathon 2.0 — 2026
Phase	Phase 2: Build Sprint — Prototype Submission
Problem Statement	PS5: AI-Powered Urban Air Quality Intelligence for Smart City Intervention
Theme	Smart Cities / Environmental Intelligence / Geospatial Analytics / Public Health
Team Leader	Bala Karnayana
Institution	Mohan Babu University, Tirupati (B.Tech CSE, 2027)
GitHub	github.com/karnayanaBala/et-ai-hackathon-ps5-airquality

Criteria	Weight	Our Approach
Innovation	25%	Multi-agent AI fusing satellite + IoT + weather for hyperlocal forecasts
Business Impact	25%	Direct NCAP alignment; enforcement reduces response time hours→minutes
Technical Excellence	20%	LSTM + RF ensemble; AERMOD dispersion; RAG multilingual advisory
Scalability	15%	Kafka pipeline; API-first; deployable across all 132 NCAP cities
User Experience	15%	Dual interface (admin + citizen); 12 languages; mobile-first

1. Executive Summary

VayuDrishhti is an AI-powered Urban Air Quality Intelligence platform designed to transform how Indian city administrations monitor, predict, and respond to air pollution. Built for ET AI Hackathon 2026 under Problem Statement 5, it addresses a critical gap: despite India deploying 900+ CAAQMS monitoring stations under the National Clean Air Programme, only 31% of cities have actionable multi-agency response protocols linked to those readings.

The platform fuses real-time IoT sensor data, Sentinel-5P satellite imagery, IMD meteorological forecasts, traffic mobility feeds, and geospatial land use layers into a unified intelligence system — enabling proactive, evidence-based intervention rather than reactive dashboarding.

Feature	Capability	Technology
Ward-Level AQI Map	Real-time geospatial heatmap across all city wards	PostGIS + Leaflet.js
24–72hr AI Forecast	Hyperlocal 1km grid AQI prediction	LSTM + Random Forest Ensemble
Source Attribution	Ward-level polluter identification with confidence scores	Random Forest + AERMOD Dispersion
Enforcement Intelligence	AI-ranked action priorities with auto evidence packages	Multi-Agent AI + RAG
Citizen Advisory	Real-time health alerts in 12 Indian languages	LLM (Claude API) + IVR
Multi-City Dashboard	Compare 5+ cities: Delhi, Mumbai, Bengaluru...	React + Chart.js

2. Problem Statement & Context

India's Air Quality Crisis — Key Facts:

Metric	Data	Source
Premature deaths from air pollution	1.67 million/year	Lancet Planetary Health
Indian cities in top 50 most polluted	24 of 50 are Tier 1/2 cities	CPCB 2024
Delhi average AQI (2024-25)	218 — Poor/worse 200+ days	CPCB CAAQMS
CAAQMS stations deployed	900+ under NCAP	Ministry of Environment
Cities with actionable protocols	Only 31%	CAG Audit 2024
Mumbai hazardous AQI days (2024)	60+ days	CPCB

Core Problem:

City administrations need more than dashboards. They need: (1) Geospatial attribution — which sources are responsible at this location, right now; (2) Predictive forecasting — what will AQI be in 24 hours at ward level; and (3) Enforcement intelligence — where to deploy inspectors for maximum impact. This combination does not exist today.

3. System Architecture

VayuDrishhti is built as a 5-layer intelligence stack:

Layer 1 — Data Sources

- CAAQMS Sensors: 900+ real-time IoT monitoring stations (CPCB/NCAP network)
- Sentinel-5P Satellite: NO₂, SO₂, CO, aerosol optical depth at 5.5km resolution
- IMD Weather API: Wind vectors, temperature, humidity, boundary layer height
- Traffic & Mobility: Vehicle counts, AIS vessel data, GPS fleet tracking
- Permit Registries: Construction site permits, industrial emission clearances

Layer 2 — Data Ingestion & Processing

- Apache Kafka: High-throughput real-time stream processing of sensor feeds
- Data Quality Control: Outlier detection, sensor calibration, gap-filling algorithms
- TimescaleDB: Time-series optimized database for historical and real-time data
- PostGIS / GeoPandas: Geospatial feature extraction, ward boundary mapping

Layer 3 — AI / ML Models

- LSTM Forecasting: Recurrent neural network for 24-72hr AQI time-series prediction at 1km grid
- Random Forest: Ensemble model for source attribution (vehicle, industrial, construction, burning)
- AERMOD Dispersion Model: Atmospheric dispersion modelling for plume trajectory prediction
- LLM Advisory Engine: Claude/GPT-4 for generating multilingual citizen health advisories

Layer 4 — Intelligence Agents

- Source Attribution Agent: Ward-level polluter identification with statistical confidence scores
- Forecast Agent: 1km grid, 72hr prediction with uncertainty quantification
- Enforcement Agent: Priority-ranked action recommendations + auto-generated evidence packages
- Alert Orchestrator: Multi-channel alert dispatch (app, IVR, SMS, command center)

Layer 5 — Output Interfaces

- VayuDrishhti Command Dashboard: React-based real-time geospatial AQI maps and analytics
- Citizen Mobile App: Push notifications, ward-level advisories in 12 Indian languages
- Enforcement Reports: Auto-generated PDF evidence packages for pollution control boards
- Open API: REST API for integration with smart city platforms and CPCB systems

4. AI & ML Model Details

4.1 AQI Forecasting — LSTM + Random Forest Ensemble

The forecasting engine uses a two-model ensemble approach:

- **LSTM (Long Short-Term Memory):** Captures temporal dependencies in AQI time-series across 168-hour lookback windows. Input features: historical AQI, PM2.5, PM10, NO2, SO2, meteorological variables (wind speed/direction, temperature, humidity, boundary layer height). Output: 24-72hr AQI at 1km grid resolution.
- **Random Forest Regressor:** Handles non-linear relationships between pollution sources and observed AQI. Features include: traffic density, construction permit counts, industrial activity indices, satellite-derived aerosol data, seasonal emission calendars.
- **Ensemble:** Weighted average of both models with confidence intervals. RMSE target: <15 µg/m3 for PM2.5 24hr forecast.

4.2 Pollution Source Attribution

Source attribution identifies what fraction of observed pollution comes from each source category at ward level:

Source Category	Detection Method	Data Inputs
Vehicle Emissions (38%)	Traffic correlation + NOx/CO ratios	GPS fleet data, traffic sensors, AIS
Industrial Stack (28%)	Plume dispersion back-trajectory	SCADA emissions, satellite thermal anomalies
Construction Dust (19%)	PM10/PM2.5 ratio + permit overlay	Construction permits, satellite dust maps
Waste Burning (15%)	CO spike + thermal detection	Sentinel-5P CO, MODIS thermal

4.3 Multilingual Advisory Generation (LLM)

When AQI thresholds are breached, the LLM advisory engine:

- **Receives:** current AQI value, pollutant breakdown, ward demographics (hospital density, school density, elderly population %)
- **Generates:** personalized health advisories calibrated to risk level and vulnerable population density
- **Translates:** into English, Telugu, Hindi, Kannada, Tamil, Bengali, Marathi, Gujarati, Punjabi, Odia, Malayalam, Assamese
- **Delivers:** via mobile push, IVR call, WhatsApp, public display boards, and municipal SMS systems

5. Working Prototype — VayuDrishti Dashboard

The working prototype (index.html) demonstrates the full intelligence platform with simulated real-time data:

Feature	Description	Status
Multi-City Selection	Switch between Delhi, Mumbai, Bengaluru, Kolkata, Chennai with live AQI data	LIVE
Ward-Level AQI Map	SVG geospatial map showing 10+ wards with color-coded AQI circles	LIVE
Hero Stats Dashboard	City AQI, PM2.5, temperature, humidity, ward count — real-time updates	LIVE
24-Hour Forecast Chart	Chart.js line graph showing actual vs AI-predicted AQI trajectory	LIVE
Source Attribution Grid	Visual breakdown of pollution sources with percentage contribution	LIVE
Pollutant Levels	PM2.5, PM10, NO2, SO2, CO, O3 with WHO safe limit indicators	LIVE
Active Alert System	Critical/warning/info alerts with timestamps and recommended actions	LIVE
Multilingual Advisory	4-language toggle: English, Telugu, Hindi, Kannada	LIVE

Enforcement Priorities	AI-ranked enforcement targets with priority classification	LIVE
Responsive Design	Mobile and desktop optimized dark-theme interface	LIVE

6. Business Impact & Scalability

Direct Impact Metrics:

Impact Area	Current State	With VayuDrishti	Improvement
Pollution Source Detection	Manual inspection (days)	Real-time attribution (minutes)	1000x faster
Enforcement Response Time	48-72 hours avg	Signal to action < 2 hours	24-36hr saved/incident
AQI Forecast Lead Time	0 (reactive only)	24-72hr advance warning	Proactive intervention
Citizen Alert Coverage	English SMS only	12 languages, 4 channels	12x language coverage
Cities Deployable	Pilot only	All 132 NCAP cities via API	National scale

Scalability Architecture:

- Kafka-based ingestion pipeline horizontally scales to 10,000+ sensor endpoints
- TimescaleDB supports petabyte-scale time-series with automatic data tiering
- Microservices architecture: each AI agent independently deployable and scalable
- REST API with OpenAPI spec enables integration with Smart City Mission platforms, CPCB SAMEER app, and municipal ERP systems
- Multi-tenant SaaS model: single platform can serve all 132 NCAP cities with city-specific configurations
- Edge deployment option: lightweight sensor agents can run on CAAQMS hardware for offline resilience

7. Team Profile

Team Leader: Bala Karnayana

Attribute	Detail
Name	Bala Karnayana
Education	B.Tech CSE — Mohan Babu University, Tirupati (2027) CGPA: 80%
Current Internships	AI/ML Intern @ SmartBridge (GenAI, Agentblazer) Data Analytics Intern @ ApexPlanet
Hackathon Experience	HackIndia ChainForge 2026 (Web3) Cognizant Technoverse 2026 ChainBreakers
Certifications	Google AI Professional Microsoft GenAI & Ethics L&T Deep Learning Engineer TATA GenAI
Key Skills	Python, SQL, JavaScript, HTML/CSS, Data Analysis, Blockchain Basics, AI/ML, GenAI
GitHub	github.com/karnayanaBala
NCET Score	75.50% Grade A+ (Excellent) — May 2026

8. Future Roadmap

Phase	Timeline	Deliverable
Phase 1 — Prototype	Hackathon (Done)	Working dashboard with mock data, architecture, multilingual advisory
Phase 2 — MVP	Month 1-2	Live CPCB CAAQMS API integration, trained LSTM model on 2-year historical data, 1 city deployment
Phase 3 — City Rollout	Month 3-6	5 city deployment, mobile app launch, enforcement dashboard for PCBs, IMD weather integration
Phase 4 — National Scale	Month 6-12	All 132 NCAP cities, CPCB SAMEER integration, policy simulation module, satellite real-time validation
Phase 5 — AI Enhancement	Year 2	Digital twin for city air quality simulation, autonomous enforcement escalation, predictive analytics

